

Customer Churn Prediction

Capstone Task — Mini ML Use Case | InternSpark

Objective

Predict which customers are likely to churn using account and service data, so the business can target retention efforts at the customers most at risk instead of treating everyone the same.

Data

2,000 customer records with 13 features covering tenure, contract type, billing, internet/support services, and demographics. Churn is imbalanced at roughly 25% of customers, which shaped how the models were evaluated.

Approach

- Explored churn patterns against tenure, contract type, monthly charges, and payment method
- Cleaned a small number of missing values in TotalCharges and encoded categorical features
- Trained and compared three models: Logistic Regression, Random Forest, and Gradient Boosting, using balanced class weights to account for the churn imbalance
- Evaluated with accuracy, precision, recall, F1, and AUC rather than accuracy alone, since recall on churners matters more than overall accuracy for this problem

Results

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	0.72	0.46	0.78	0.58	0.81
Random Forest	0.75	0.51	0.24	0.33	0.75
Gradient Boosting	0.80	0.66	0.38	0.48	0.80

Logistic Regression was selected as the best model on AUC (0.81) and, more importantly, recall (0.78). It catches far more of the customers who actually churn than the tree-based models, which is the more useful trade-off for a retention use case even though its precision is lower.

Key Drivers of Churn

- Contract type — month-to-month customers churn substantially more than those on annual contracts
- Tenure — churn risk is highest in the first several months of the relationship
- Monthly charges — higher bills correlate with higher churn
- Support add-ons and payment method — customers without tech support/security add-ons, and those paying by electronic check, churn more than other segments

Recommendation

Use the model's churn probability to flag the highest-risk 15–20% of active customers each month and route them to a retention team, with a particular focus on new, month-to-month customers on higher price plans. Incentivizing longer contracts and improving the visibility of support add-ons at signup are the two most direct levers for reducing churn.